BESS HAGAN

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SUMMARY

Computer Science graduate with strong foundations in Python, SQL, and data analysis. Experience designing and implementing a SQL-to-Python data pipeline, performing exploratory analysis, and transforming raw data into actionable insights through research and applied projects. Seeking entry-level data analyst or analytics-focused roles.

EDUCATION

Bachelor of Arts in Computer Science, *Minor in Data Analytics* 12/2025
Southwestern University, Georgetown, TX GPA: 4.0, *Summa Cum Laude*

- **Relevant Coursework:** Data Analytics, Database Management, Machine Learning, Algorithms

Associate of Science in Computer Science 05/2023
Temple College, Temple, TX GPA: 4.0

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, R
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Excel (Pivot Tables, Dashboards), Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing
Databases & Tools: PostgreSQL, pgAdmin, Jupyter Notebook, Anaconda, GitHub, VS Code
Machine Learning (Applied): Linear & Logistic Regression, Decision Trees, KNN, K-Means Clustering, scikit-learn, PyTorch

EXPERIENCE

Student Researcher 05/2024 – 08/2024
DREU Program, INVITE Institute, University of Illinois Urbana-Champaign, Champaign, IL

- Built a SQL-to-Python data pipeline to extract, transform, and analyze student interaction logs from a database backed virtual learning environment.
- Identified and diagnosed data quality and logging issues between the application and database layers, collaborating with developers to improve data consistency and reliability.
- Developed Python scripts to compute and summarize student interaction metrics in support of statistical analysis of engagement and persistence.

Student Researcher, SURF 2025 (Procedural Content Generation) 05/2025 – 07/2025
Southwestern University, Georgetown, TX

- Trained and evaluated generative models in Python using PyTorch, implementing early stopping, checkpointing, and metric tracking to support reproducible experimentation.
- Developed tooling to collect, log, and visualize model evaluation metrics, enabling systematic validation of experimental results.
- Coauthored a peer-reviewed paper accepted to the 21st AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE-25).

Student Software Engineer, Capstone: Senior Seminar in Software Engineering 01/2025 – 05/2025
Southwestern University, Georgetown, TX

- Collaborated with an Agile team using GitHub for version control, feature branching, and issue tracking to develop an educational game in the Godot Engine.
- Implemented a drag-and-drop mini-game with a state-driven tutorial system and dynamic validation logic, integrating a YAML-based dialogue manager and tween animations for user guidance and feedback.

LEADERSHIP ROLES

- **Chapter President**, Upsilon Pi Epsilon National Honorary Computer Science Society 08/2024 – 05/2025
- **Vice President**, Lambda Theta Chapter, Phi Theta Kappa Honor Society 07/2022 – 05/2023